Time integration and adjoint-graph differentiation for reflected-clock Brownian observables

Project Stardust

Reviewed editorial draft, October 30, 2026

Abstract

For independent Brownian motions B, W, let $Z_t = B(|W_t|)$ and $A_T = \int_0^T Z_s\,ds$. Although Z_T is outside the original clock Gaussian Sobolev domain at every positive time, A_T belongs to the intersection of the two original Gaussian derivative domains. A folded Gaussian transition bound gives a uniform finite clock energy for spatially mollified averages, and closedness gives the actual average-domain result. Bounded smooth functions of averages form a dense derivative domain in each fixed raw observed past. For the resulting closed derivative and its maximal Hilbert adjoint, the embedded gradient of A_T is differentiable in adjoint-graph-dual norm, including left differentiation at the past endpoint. Its derivative pairs with actual Z_T but has no native field representative. Along every vanishing positive or negative time-step sequence, the native gradient-slope norms diverge. At a fixed horizon, contractions with each fixed clock direction have actual spatial occupation kernels whose mollifications converge strongly; full-vector strong convergence for those linear approximants is not inferred. Separately, the nonlinear sine time average belongs to the same original joint domain, and both its native gradient channels converge strongly in full Hilbert norm. A compact spatial operator bound, summable occupation trace and uniform cutoff tail prove the vector limit. These results give precise source-valued operator identities, not an observed-filtration Itô or Clark–Ocone formula. They reconstruct frozen formal results; novelty remains unresolved.

1 The process, operator and statement

Let B be a two-sided standard Brownian scenery and W an independent one-sided standard Brownian motion, on their product Gaussian probability space $(\Omega, \mathcal{F}, P)$. Put

$$Z_t = B(|W_t|), \qquad A_T = \int_0^T Z_s\,ds \quad (t, T \geq 0).$$

Only the positive scenery branch is used by Z. Extending it independently to negative arguments permits spatial convolution. This is the simple Brownian-time Brownian model of [1]; an excursion does not select a fresh scenery, and the clock is reflected. The signed-clock process $B(W_t)$ is a different model.

Let D_B, D_W be the original closed Gaussian gradients on the product space, acting respectively in the scenery and clock factors. Their coefficient Hilbert spaces can be realized as $H_B = L^2(\mathbb{R})$ and $H_W = L^2(\mathbb{R}_+)$, using white noise. Equivalently one can use the frozen library's isometric separable Gaussian coordinates. Additional unused independent coordinates tensor the graph with constants; averaging over them gives the reverse domain implication. All fields below retain the ambient Gaussian codomain

$$\mathcal{G} = L^2(P; H_B) \oplus L^2(P; H_W).$$

They are not assumed observable or predictable in physical time.

Set $E = L^2(\sigma(Z_s : s \geq 0), P)$, understood as the closed subspace of $L^2(P)$ with measurable representatives, and define the *native* operator

$$\mathcal{D}f = (D_B f, D_W f), \qquad D(\mathcal{D}) = E \cap D(D_B) \cap D(D_W). \tag{1}$$

The domain is the entire stated intersection. For fixed $H \geq 0$ let

$$\mathcal{F}_H^Z = \sigma(Z_s : 0 \leq s \leq H), \qquad E_H = L^2(\mathcal{F}_H^Z, P), \qquad \mathcal{D}_H = \mathcal{D}|_{E_H}.$$

This is the raw observed past; L^2 always identifies almost-everywhere equal variables. Taking that quotient does not assert right continuity of the filtration.

For a closed densely defined operator $T : E_0 \supset D(T) \to \mathcal{G}$, write

$$\mathcal{T}_T = D(T^*), \quad \|v\|_{\mathcal{T}_T}^2 = \|v\|_{\mathcal{G}}^2 + \|T^* v\|_{E_0}^2,$$

and define continuous linear functionals

$$(j_T V)(v) = \langle V, v \rangle, \qquad (d_T f)(v) = \langle f, T^* v \rangle. \tag{2}$$

The adjoint domain here is maximal: $v \in D(T^*)$ exactly when $f \mapsto \langle v, Tf \rangle$ is bounded in the E_0 norm on $D(T)$.

Theorem 1.1. *The operators $\mathcal{D}$ and $\mathcal{D}_H$ are closed and densely defined. For every $T \geq 0$, $A_T \in D(\mathcal{D})$; if $T \leq H$, also $A_T \in D(\mathcal{D}_H)$ and $\mathcal{D}_H A_T = \mathcal{D} A_T$. Write $V_T = \mathcal{D} A_T$. For every $T > 0$,*

$$\frac{d}{dT} j_{\mathcal{D}} V_T = d_{\mathcal{D}} Z_T \quad \text{in } \mathcal{T}_{\mathcal{D}}' \text{ norm.} \tag{3}$$

For $0 < T \leq H$ and every $h_n < 0$ with $h_n \to 0$,

$$j_{\mathcal{D}_H} \frac{V_{T+h_n} - V_T}{h_n} \longrightarrow d_{\mathcal{D}_H} Z_T \quad \text{in } \mathcal{T}_{\mathcal{D}_H}' \text{ norm,} \tag{4}$$

where the finitely many terms with $T + h_n < 0$ may be omitted. The limit is nonzero and has no representative $j_{\mathcal{D}_H} V$ with $V \in \mathcal{G}$. Moreover

$$\left\| \frac{V_{T+h_n} - V_T}{h_n} \right\|_{\mathcal{G}} \longrightarrow \infty. \tag{5}$$

The full operator has the same nonrepresentability, and (5) also holds for every positive vanishing step sequence. For each $v \in D(\mathcal{D}_H^*)$ the fixed-past derivative has the actual pairing

$$(d_{\mathcal{D}_H} Z_T)(v) = \mathbb{E}[Z_T \, \mathcal{D}_H^* v]. \tag{6}$$

The proof separates average-domain membership, density on the observed past, and the change of topology needed for time differentiation. None of these steps assumes a Gaussian law for Z or a predictable representation property for its observed filtration.

2 Time continuity and scenery differentiation

Let $e_r = \mathbf{1}_{(0,r]} \in H_B$ for $r \geq 0$. The scenery white-noise isometry sends e_r to $B(r)$, and

$$\|B(r) - B(q)\|^2_{L^2(B)} = |r - q| \quad (r, q \geq 0).$$

Conditioning on the entire clock gives

$$\|Z_t - Z_s\|_2^2 = \mathbb{E}||W_t| - |W_s|| \leq \mathbb{E}|W_t - W_s| = \sqrt{2/\pi}\,|t - s|^{1/2}. \tag{7}$$

Consequently Z is continuous as an $L^2(P)$-valued curve. It is Bochner integrable on compact time intervals, and the integral agrees almost surely with the integral of its continuous paths, by Fubini. The fundamental theorem of calculus gives

$$A'_T = Z_T \quad \text{in } L^2(P), \qquad T > 0. \tag{8}$$

The same derivative holds in the closed subspaces E and, from the left for $T \leq H$, E_H.

For fixed clock sample, $B(|W_t|)$ is first Gaussian chaos in B, with scenery gradient $e_{|W_t|}$. This gives the product closed graph and

$$\|D_B Z_t\|^2_{L^2(P;H_B)} = \mathbb{E}|W_t| = \sqrt{2t/\pi}.$$

Here graph membership is genuine: approximate e_r by finite orthogonal projections in the scenery coefficient space, and use dominated convergence with majorant $|W_t|$ for the squared errors. The finite linear Gaussian graphs converge strongly to the stated product graph. Integrating that graph yields

$$A_T \in D(D_B), \qquad D_B A_T = \int_0^T e_{|W_s|}\, ds. \tag{9}$$

Indeed both value and gradient are Bochner integrable: the gradient norm is a constant times $s^{1/4}$, which is integrable on $[0, T]$. A closed linear graph contains the integral of any integrable curve in that graph, by approximation with simple graph-valued functions.

3 The folded transition estimate and average clock domain

Choose the frozen nonnegative smooth probability mollifiers ρ_n, supported in $[-\epsilon_n, \epsilon_n]$, $\epsilon_n = 2/(n+1)$, and let

$$B_n(r) = \int \rho_n(u) B(r - u)\, du, \quad Z_t^{(n)} = B_n(|W_t|), \quad A_T^{(n)} = \int_0^T Z_s^{(n)}\, ds.$$

The Brownian increment metric and Minkowski's inequality give, uniformly in $r \in \mathbb{R}$,

$$\|B_n(r) - B(r)\|_{L^2(B)} \leq \int \rho_n(u) \sqrt{|u|}\, du \leq \sqrt{\epsilon_n}.$$

Independence therefore gives

$$\|A_T^{(n)} - A_T\|_2 \leq T\sqrt{\epsilon_n}. \tag{10}$$

These mollified approximants need not be observed functions of Z; they are used only in the original ambient clock graph.

Put $q_s = \mathbf{1}_{(0,s]} \in H_W$ and $\operatorname{sgn}(0) = 0$. In the original product Gaussian clock domain,

$$D_W Z_s^{(n)} = B_n'(|W_s|)\operatorname{sgn}(W_s) q_s, \qquad G_T^{(n)} = D_W A_T^{(n)} = \int_0^T D_W Z_s^{(n)}\, ds. \tag{11}$$

To justify the point graph without assuming the scenery path has a classical derivative before mollification, expand $B_n(r)$ in a scenery orthonormal basis. Its scalar coefficients have derivative in the original one-dimensional Gaussian graph after $r = |W_s|$. The absolute value has weak derivative sgn; approximating it by $\sqrt{x^2+\delta^2}$ and then taking the Gaussian closed graph proves that assertion. For the scenery sums, the square norms of the derivative coefficients sum to $\|\rho_n\|_2^2$, uniformly in r. Thus the value and derivative sums converge in the product L^2 graph. At $s = 0$ the random value is clock independent and the derivative is zero. On $[0,T]$ the derivative norms are bounded by $\sqrt{s}\|\rho_n\|_2$, so graph integration gives the second equality in (11).

Lemma 3.1. *For every n and $T \geq 0$,*

$$\|G_T^{(n)}\|^2_{L^2(P;H_W)} \leq \frac{8}{\sqrt{2\pi}} T^{5/2}. \tag{12}$$

Proof. By the scenery white-noise isometry,

$$\mathbb{E}_B[B_n'(r)B_n'(q)] = k_n(r,q) := \int_{\mathbb{R}} \rho_n(r-x)\rho_n(q-x)\, dx \geq 0.$$

For fixed r, $\int_{\mathbb{R}} k_n(r,q)\, dq = 1$ by Tonelli and normalization. For $0 \leq s < t$, conditional on $W_s = y$, the density of $|W_t|$ on $q > 0$ is

$$p_{t-s}(q-y) + p_{t-s}(-q-y) \leq \frac{2}{\sqrt{2\pi(t-s)}}.$$

Hence $\mathbb{E}k_n(|W_s|,|W_t|) \leq 2/\sqrt{2\pi(t-s)}$. In the double integral for the squared norm of (11), the coefficient inner product is $\langle q_s, q_t\rangle = \min(s,t)$. After scenery averaging, the signs have absolute value at most one and $k_n \geq 0$. Thus off the time diagonal the absolute integrand is bounded by

$$\frac{2T}{\sqrt{2\pi}}|t-s|^{-1/2}.$$

For each fixed mollifier, its actual diagonal covariance is finite and is not assigned zero. The diagonal is a product-Lebesgue null set, so it does not affect integration. The bound is integrable, since

$$\int_0^T |t-s|^{-1/2}\, dt = 2\sqrt{s} + 2\sqrt{T-s} \leq 4\sqrt{T}.$$

Integrating over $s \in [0,T]$ proves (12). The same domination justifies the double integrals and conditioning. The constant is a convenient uniform estimate, not a sharp energy formula. □

Proposition 3.2. *The actual average A_T belongs to $D(\mathcal{D})$ for every $T \geq 0$. Its original clock gradient satisfies $\|D_W A_T\|_2^2 \leq 8T^{5/2}/\sqrt{2\pi}$.*

Proof. For $T = 0$ the result is immediate. Otherwise (10) gives strong value convergence, and Lemma 3.1 bounds the actual clock gradients. A bounded sequence in this Hilbert field space has a weakly convergent subsequence, say $G_T^{(n_k)} \rightharpoonup G$. The graph of the closed linear operator D_W is a norm-closed subspace, and hence is weakly closed: a point outside it can be separated

by its orthogonal projection. Therefore $(A_T^{(n_k)}, G_T^{(n_k)}) \rightharpoonup (A_T, G)$ lies in that graph. It follows that $A_T \in D(D_W)$, and lower semicontinuity of the norm gives the estimate. Equation (9) gives scenery-domain membership. Finally A_T is observable from Z and is an element of E, which proves (1) membership. □

Uniform derivative bounds alone have been used for membership, not to assert strong gradient convergence. The library has a further strong directional occupation-kernel convergence theorem, proved separately below; it is unnecessary for the time-differentiation result and does not assert full-vector strong convergence.

4 Density in the fixed raw observed past

Closedness of $\mathcal{D}$ follows from closedness of both partial graphs and of E. Its restriction $\mathcal{D}_H$ is closed because E_H is closed. Density requires proof; it does not follow merely from restricting a dense ambient domain.

Lemma 4.1. *Finite linear combinations of A_t, $0 \le t \le H$, belong to $D(\mathcal{D}_H)$. Their sine and cosine also belong to that domain.*

Proof. Linearity and Proposition 3.2 prove the first assertion; all averages up to H are $\mathcal{F}_H^Z$-measurable. For either Gaussian factor, the closed gradient satisfies the bounded C^1 chain rule $D\phi(f) = \phi'(f)Df$ when ϕ, ϕ' are bounded and ϕ' is continuous. To see this from the original closure, take smooth cylindrical graph approximants $f_n \to f$, $Df_n \to Df$. Lipschitz continuity gives $\phi(f_n) \to \phi(f)$ in L^2. The derivative products converge: the part containing $Df_n - Df$ is bounded by $\|\phi'\|_\infty \|Df_n - Df\|_2$, and the part containing $(\phi'(f_n) - \phi'(f))Df$ tends to zero by convergence in probability and bounded dominated convergence. A subsequence argument gives the same limit for the whole sequence. Smooth compositions in finitely many Gaussian coordinates belong to the original closure: cut off the coordinates, identify the weak partial derivatives by Gaussian integration by parts, and expand in tensor Hermite polynomials. The derivative coefficient identities and Parseval show that polynomial partial sums converge with all first derivatives in L^2. Cutoff errors vanish by Gaussian integrability and the bounded derivative. Thus the finite-dimensional chain identities hold in the original closed graph, and closedness proves the asserted chain rule. Apply it in each factor to sine and cosine. Their compositions remain observable in $\mathcal{F}_H^Z$. □

Proposition 4.2. *$D(\mathcal{D}_H)$ is dense in E_H, including at $H = 0$, and $D(\mathcal{D})$ is dense in E.*

Proof. For $0 < t \le H$ and $0 < \epsilon < t$, put $Q_{t,\epsilon} = (A_t - A_{t-\epsilon})/\epsilon$. By (7), $Q_{t,\epsilon} \to Z_t$ in L^2 as $\epsilon \downarrow 0$. Thus Lemma 4.1 and Lipschitz continuity show that the sine and cosine of every finite linear combination of points $Z_{t_1}, \dots, Z_{t_m}$ lie in the L^2 closure of $D(\mathcal{D}_H)$. Zero-time points cause no difficulty since $Z_0 = 0$. Constants are in the domain with zero gradient.

Suppose $g \in E_H$ is orthogonal to $D(\mathcal{D}_H)$. It is integrable by Cauchy–Schwarz. For each finite observation vector $X = (Z_{t_1}, \dots, Z_{t_m})$, the finite signed measure

$$\lambda(S) = \mathbb{E}[g\mathbf{1}_{\{X \in S\}}]$$

has all Fourier transforms zero, by the preceding sine/cosine approximation. Fourier uniqueness implies $\lambda = 0$. One can verify uniqueness by convolving λ with a centered Gaussian density:

Fourier inversion for the Gaussian and Fubini make each such convolution zero; Gaussian approximate identities then recover the integrals of compactly supported continuous functions, and hence the finite measure. Therefore $\mathbb{E}[g\mathbf{1}_A] = 0$ for every finite observation cylinder A.

These cylinders form a pi-system generating the raw past. The signed measure $A \mapsto \mathbb{E}[g\mathbf{1}_A]$ has total mass zero; the pi-lambda theorem makes it zero on the whole generated sigma-algebra. Since g has a past-measurable representative, taking its positive and negative sets gives $\mathbb{E}|g| = 0$. Hence $g = 0$. The orthogonal complement is zero, which proves density. When $H = 0$, E_0 consists of constants and the conclusion is immediate. The same argument with arbitrary finitely many nonnegative times proves density in E. □

This proof uses the actual finite observation laws; it does not replace them with Gaussian laws or identify conditional expectation with a source-chaos projection.

5 Why positive-time points remain outside the domain

For clarity we recall the clock obstruction underlying the time result, with its common-field requirement. In the scenery-valued Hilbert space $K = L^2(B)$, for $c > 0$ consider $F_c(x) = B(c|x|)$. For $0 \le a < x$,

$$\|F_c(x) - F_c(a)\|_K^2 = c(x - a). \tag{13}$$

If F_c had a locally square-integrable Bochner weak derivative g, its continuous representative would satisfy $F_c(x) - F_c(a) = \int_a^x g(u)\,du$. Cauchy–Schwarz and (13) would give

$$c \le \int_a^x \|g(u)\|_K^2\,du.$$

The right side tends to zero as $x \downarrow a$, a contradiction.

Lemma 5.1. *For every $T > 0$, $Z_T \notin D(D_W)$ and therefore $Z_T \notin D(\mathcal{D})$. If $T \le H$, also $Z_T \notin D(\mathcal{D}_H)$.*

Proof. Set $G = W_T/\sqrt{T}$, a standard Gaussian coordinate, and $h = \mathbf{1}_{(0,T]}/\sqrt{T} \in H_W$. Decompose the remaining clock Gaussian coordinates orthogonally to h. If a full original clock gradient V existed, contract it with h and conditionally average over those remaining clock coordinates. Jensen gives a common field $g \in L^2(\gamma; K)$, where γ is standard Gaussian measure. The product Gaussian integration-by-parts identity, tested against a scenery variable $e \in K$ and a compactly supported smooth scalar $p(G)$, gives

$$\int \langle F_{\sqrt{T}}(x), e\rangle (p'(x) - xp(x))\,d\gamma(x) = -\int \langle g(x), e\rangle p(x)\,d\gamma(x).$$

This identity follows first on the Gaussian core and passes to the closed graph by Cauchy–Schwarz. The recovered g is the same field for every e, a fact essential to the conclusion.

Let $\gamma(x)$ also denote the strictly positive Gaussian density. Substitute $p = \psi/\gamma$ on any compact interval. Since $\gamma' = -x\gamma$, one has $(p' - xp)\gamma = \psi'$. The displayed identity becomes the distributional weak derivative identity relative to Lebesgue measure. On that interval g is square integrable because the Gaussian density is bounded below. Testing a countable dense set of scenery vectors shows that $F_{\sqrt{T}}(x) - \int_a^x g(u)\,du$ is one constant K-valued vector almost everywhere. Indeed every scalar projection has zero weak derivative; separability and a common full-measure set identify that vector. Both sides have continuous representatives, so the equality holds throughout the interval. This contradicts (13). The two native exclusions follow from their domain inclusions. □

This is the point obstruction from the companion [4]. The scenery gradient in Section 2 remains valid; a clock chain rule is not applied to a point outside its domain.

6 Adjoint-graph duals and time-slope escape

We give the closed-operator facts explicitly to fix the test space. For $T : E_0 \supset D(T) \to \mathcal{G}$ closed and densely defined, the maps in (2) satisfy

$$\|d_T f\|_{\mathcal{T}_T'} \le \|f\|_{E_0}, \quad j_T(Tu) = d_T u, \quad \|j_T(Tu) - d_T f\|_{\mathcal{T}_T'} \le \|u - f\|_{E_0}. \tag{14}$$

The first bound is Cauchy–Schwarz and $\|T^* v\| \le \|v\|_{\mathcal{T}_T}$; adjointness gives the equality, and applying the first bound to $u - f$ gives the last estimate.

Lemma 6.1. *The map j_T is injective, and $d_T f = j_T V$ for some $V \in \mathcal{G}$ if and only if $f \in D(T)$, in which case $V = Tf$. If $u_n \in D(T)$ converges to $f \notin D(T)$ in E_0, then $\|Tu_n\|_{\mathcal{G}} \to \infty$.*

Proof. The orthogonal complement of the graph is

$$\Gamma(T)^{\perp} = \{(-T^* v, v) : v \in D(T^*)\}.$$

This follows directly by testing $\langle x, u\rangle + \langle y, Tu\rangle = 0$ on $u \in D(T)$; density makes $x = -T^* y$ unique. The relation $d_T f = j_T V$ says precisely that (f, V) is orthogonal to this complement. Since the graph is closed, $(f, V) \in \Gamma(T)$. The converse is adjointness. Taking $f = 0$ proves injectivity of j_T.

If $\|Tu_n\|$ did not tend to infinity, a subsequence would have bounded outputs and hence a further weakly convergent subsequence, say to V. Then (u_n, Tu_n) along that subsequence converges weakly to (f, V). The closed linear graph is weakly closed, so $f \in D(T)$, a contradiction. This proves full norm escape, rather than mere unboundedness of one subsequence. □

Proof of Theorem 1.1. Propositions 3.2 and 4.2, together with closedness already noted, give the operator assertions. Put $V_t = \mathcal{D} A_t$. Adjointness gives $j_{\mathcal{D}} V_t = d_{\mathcal{D}} A_t$. The bounded linear map $d_{\mathcal{D}}$ commutes with the derivative (8), proving (3). For fixed H, the same equality holds with $\mathcal{D}_H$ for $0 \le t \le H$. The left value quotients converge in E_H to Z_T even when $T = H$. Applying the bound in (14) proves (4); more precisely, its error is at most

$$\left\| \frac{A_{T+h_n} - A_T}{h_n} - Z_T \right\|_2 \longrightarrow 0.$$

By Lemmas 5.1 and 6.1, $d_{\mathcal{D}_H} Z_T$ is not represented by any $j_{\mathcal{D}_H} V$. It cannot be zero, because zero has representative $V = 0$. The full-operator conclusion is identical.

The value quotients $u_n = (A_{T+h_n} - A_T)/h_n$ lie in $D(\mathcal{D})$ and converge to $Z_T \notin D(\mathcal{D})$. Their gradients are $(V_{T+h_n} - V_T)/h_n$, so Lemma 6.1 proves (5). The proof applies to positive steps as well. In particular V_t has no strong ordinary-time derivative at any positive T: such a derivative would force these slope norms to remain bounded. Finally (6) is the definition of $d_{\mathcal{D}_H}$ as an L^2 pairing. Its integrand is integrable by Cauchy–Schwarz. □

No tests or past spaces change with the step index in (4). This is differentiation on each *fixed* raw observed past. It does not resolve convergence of fields with a changing past, and it is not a temporal stochastic integral identification.

7 Actual directional occupation kernels and their strong limit

The average-domain proof requires only bounded clock gradients. There is also a stronger statement for each fixed clock direction. We prove it without replacing a physical spatial kernel by a pointwise white-noise value. Fix $T \geq 0$ and $h \in H_W$, and put

$$a_h(s) = \langle h, q_s \rangle, \qquad X_{n,h} = \langle G_T^{(n)}, h \rangle_{H_W}, \qquad X_h = \langle D_W A_T, h \rangle_{H_W}.$$

Here a_h is continuous and $|a_h(s)| \leq \|h\|\sqrt{T}$ on $[0,T]$. Let $I : L^2(\mathbb{R}_x) \to L^2(P_B)$ denote the original scenery white-noise isometry, determined by the oriented Brownian interval indicators. Its range is closed. It is applied only to ordinary spatial L^2 inputs, never to a Dirac distribution. Independence gives the isometric Bochner lift

$$\mathcal{I} : L^2(P_W; L^2(\mathbb{R}_x)) \longrightarrow L^2(P_W \otimes P_B), \qquad (\mathcal{I}k)(W) = I(k(W)).$$

Fubini transports this to the original factor order. The equation is an $L^2(P_B)$ equality for almost every clock sample.

Theorem 7.1. *For every fixed h, T there is a unique $K_{h,T} \in L^2(P_W; L^2(\mathbb{R}_x))$ such that $X_h = \mathcal{I}K_{h,T}$. The finite-width kernels*

$$k_{n,h,T}(W,x) = \int_0^T a_h(s)\,\mathrm{sgn}(W_s)\rho_n(|W_s| - x)\,ds \tag{15}$$

are genuine members of that Bochner space, satisfy $X_{n,h} = \mathcal{I}k_{n,h,T}$, and obey

$$k_{n,h,T} \longrightarrow K_{h,T} \quad \text{in } L^2(P_W; L^2(\mathbb{R}_x)), \qquad X_{n,h} \longrightarrow X_h \quad \text{in } L^2(P). \tag{16}$$

For each fixed $\phi \in C_c(\mathbb{R})$,

$$\int_{\mathbb{R}} K_{h,T}(W,x)\phi(x)\,dx = \int_0^T a_h(s)\,\mathrm{sgn}(W_s)\phi(|W_s|)\,ds \quad \text{for almost every } W. \tag{17}$$

A specified countable test family has a common probability-one event.

Proof. For each fixed width, the spatial profile in (15) is an $L^2(\mathbb{R}_x)$ vector with norm $\|\rho_n\|_2$. Its coefficient is bounded by $\|h\|\sqrt{T}$. Translations of a fixed smooth compactly supported profile are continuous in spatial L^2, so the time/clock field is strongly measurable. Bochner integration and Minkowski show

$$\|k_{n,h,T}(W,\cdot)\|_2 \leq T\|h\|\sqrt{T}\,\|\rho_n\|_2.$$

Thus it defines the asserted clock-Bochner element. Moving the bounded isometry through the time integral in (11) proves $X_{n,h} = \mathcal{I}k_{n,h,T}$.

The full gradients $G_T^{(n)}$ converge weakly to $D_W A_T$. Indeed strong value convergence and uniform gradient bounds show that all weak subsequential limits belong to that closed graph and equal its unique gradient. Equivalently, adjoint-domain pairings converge directly and extend to all field tests by their density and the uniform bound. Consequently $X_{n,h} \rightharpoonup X_h$. The range of $\mathcal{I}$ is a closed linear subspace, hence weakly closed. It follows that $X_h = \mathcal{I}K_{h,T}$ for exactly one $K_{h,T}$, by isometry. Applying the adjoint of $\mathcal{I}$ also proves $k_{n,h,T} \rightharpoonup K_{h,T}$ in the physical Bochner space.

Test this weak limit against $\zeta(W)\phi(x)$, with $\zeta \in L^2(P_W)$ and $\phi \in C_c(\mathbb{R})$. Fubini in (15) gives

$$\langle k_{n,h,T}, \zeta \otimes \phi \rangle = \mathbb{E}_W \left[\zeta \int_0^T a_h(s) \operatorname{sgn}(W_s)(\rho_n * \phi)(|W_s|)\, ds \right].$$

The frozen mollifiers are even, nonnegative and have mass one. Their convolutions tend to ϕ pointwise and are bounded by $\|\phi\|_\infty$. The integrand is dominated by $|\zeta|\,\|h\|\sqrt{T}\|\phi\|_\infty$ on the finite time interval; $\mathbb{E}|\zeta| < \infty$. Dominated convergence proves the corresponding identity with $K_{h,T}$ and ϕ. Both random scalar sides belong to $L^2(P_W)$: the spatial side by Cauchy–Schwarz, the time side by its deterministic bound. Equality against all ζ proves (17). Countable intersection gives the stated family version, with no uncountable intersection of null sets.

The crucial additional identification is

$$k_{n,h,T}(W,\cdot) = \rho_n * K_{h,T}(W,\cdot) \quad \text{in spatial } L^2, \quad \text{for almost every } W. \tag{18}$$

Here are the representative details. For each fixed spatial x, apply (17) with $\phi(r) = \rho_n(x - r)$. Its spatial side is the continuous pairing of the Bochner kernel with a continuously translated L^2 profile. Its time side is the integral of a jointly measurable time/space/clock function. Both sides are jointly measurable in (x, W). Product-measure Fubini therefore converts the fixed-x almost-sure identities into an identity for almost every (x, W), then for almost every W and almost every x.

For the finite-width time integral its actual spatial L^2 representative is (15): for a continuous clock sample set $M = \sup_{s \le T} |W_s| < \infty$. The scalar integrand vanishes outside $[-\epsilon_n, M + \epsilon_n]$ and is bounded by $\|h\|\sqrt{T}\|\rho_n\|_\infty$. This compact spatial majorant is square integrable, so scalar Fubini agrees with the spatial Bochner integral. Lebesgue measure on the whole line has not been treated as finite. These observations prove (18) as an identity of actual stored L^2 classes, not merely of formal pairings.

Positive unit-mass convolution is an L^2 contraction, even for signed inputs: Jensen gives

$$|(\rho_n * u)(x)|^2 \le \int \rho_n(x - r)|u(r)|^2\, dr, \qquad \|\rho_n * u\|_2^2 \le \|u\|_2^2$$

by Tonelli and translation invariance. Consequently $\|k_{n,h,T}\| \le \|K_{h,T}\|$ in the full clock/spatial norm. Using the already proved weak convergence,

$$\|k_{n,h,T} - K_{h,T}\|^2 \le 2\|K_{h,T}\|^2 - 2\langle k_{n,h,T}, K_{h,T}\rangle \longrightarrow 0.$$

This proves the first strong limit; applying the isometry proves the second. Norms converge as well. At $T = 0$ all kernels and fields vanish. □

The named formal endpoint is
`IBMClockOccupationStrong.native_directional_tendsto`.
Its proof uses precisely the positive-convolution norm bound and weak pairing argument above. The intermediate physical, spatial representative and product-null-set modules are mapped separately.

A signed occupation density and the exact quantifiers

A further *classical human deduction* identifies a measure after fixing h, T. For each continuous clock path define the finite signed measure

$$\nu_{h,T,W}(A) = \int_0^T a_h(s) \operatorname{sgn}(W_s)\mathbf{1}_A(|W_s|)\, ds.$$

Its total variation is at most $T\|h\|\sqrt{T}$, and its support is in $[0, M]$. Choose, for every positive integer m, a countable uniformly dense family of continuous functions supported in $[-m, m]$ (rational piecewise linear functions suffice). Apply the countable-family identity (17) once to their union. On that common event, spatial Cauchy–Schwarz bounds the kernel pairing error for a uniform test approximation by $\sqrt{2m}\|K_{h,T}(W)\|_2$ times its uniform error. The time-side error is at most $T\|h\|\sqrt{T}$ times that error. Thus the identity holds for every compactly supported continuous test on the same event. Uniqueness of locally finite signed Radon measures gives $\nu_{h,T,W}(dx) = K_{h,T}(W, x)\,dx$ locally. It also forces the density to vanish outside $[0, M]$. Spatial L^2 on this bounded support implies L^1, so this is equality of finite signed measures globally. It asserts no common event for uncountably many h or T, and no positivity or jointly continuous local-time field. Classical Brownian occupation density is already established; the primary exposition of Pitman–Yor [5] supplies that context. The present proof instead identifies the particular density recovered from the original clock graph.

For completeness, two more *human consequences* make the topology precise. For any fixed finite orthonormal family $h_1, \dots, h_N$,

$$\mathbb{E}\left\|P_N(G_T^{(n)} - D_W A_T)\right\|_{H_W}^2 = \sum_{j=1}^{N} \|X_{n,h_j} - X_{h_j}\|_2^2 \longrightarrow 0.$$

Moreover convergence is uniform over h in any fixed compact subset of H_W. The maps $h \mapsto X_{n,h} - X_h$ have uniformly bounded operator norms by Lemma 3.1 and the average-domain bound. Cover the compact set by finitely many sufficiently small balls; use the fixed-direction limit at their centers and this common bound on the remaining errors. This is the usual finite-net proof, not an operator-norm convergence assertion on the whole unit ball.

Neither conclusion implies full-vector strong convergence: if (e_n) is an orthonormal family in an infinite-dimensional coefficient space and ξ is any fixed scalar random variable of L^2 norm one, the fields ξe_n have norm one while every fixed contraction tends to zero in scalar L^2. This is a logical counterexample, not a claimed nonconvergence theorem for the actual IBM gradients. The strong directional spatial limit also concerns mollifier width at fixed T; it coexists with native time-slope norm escape in Theorem 1.1. It supplies neither a point clock gradient nor physical-time observed stochastic integration.

8 A nonlinear average with strong native gradient approximation

There is a separate, stronger vector statement for

$$U_T = \int_0^T \sin(Z_t)\,dt.$$

This is not $\sin(A_T)$. Its proof needs more than the fixed-direction argument of the preceding section. Write $v_r = 1_{(0,r]}$ for $r \geq 0$ in the physical scenery realization and $q_t = 1_{(0,t]} \in H_W$. Use the original two-sided oriented extension of v_r at negative r. The mollified scenery B_n is the one fixed in Section 3. Define the full clock fields

$$V_n(T) = \int_0^T \operatorname{sgn}(W_t)\cos(B_n(|W_t|))B_n'(|W_t|)q_t\,dt \ \in L^2(P; H_W). \tag{19}$$

Each fixed-width integral is an ordinary Bochner integral in this Hilbert field space. Its actual representatives use the original scenery and clock, with their factor-order isometries retained. For

the source channel put $t_{j,n} = jT/(n+1)$ and

$$S_n(T) = \frac{T}{n+1} \sum_{j=1}^{n+1} \cos(Z_{t_{j,n}}) v_{|W_{t_{j,n}}|} \ \in L^2(P; H_B). \tag{20}$$

Theorem 8.1. *For every fixed $T \geq 0$, $U_T \in D(\mathcal{D})$, and*

$$S_n(T) \longrightarrow D_B U_T, \qquad V_n(T) \longrightarrow D_W U_T$$

strongly in the original respective Hilbert field norms. In particular $(S_n(T), V_n(T)) \to \mathcal{D}U_T$ in $\mathcal{G}$. Its clock energy satisfies

$$\|D_W U_T\|_2^2 \leq C_T, \qquad C_T = 8p_1 T^2 \sqrt{T} + 4T^3, \quad p_1 = (2\pi)^{-1/2}. \tag{21}$$

For every $v \in D(\mathcal{D}^*)$,

$$|\langle (S_n(T), V_n(T)), v \rangle - \mathbb{E}[U_T \mathcal{D}^* v]| \leq \|(S_n(T), V_n(T)) - \mathcal{D}U_T\|_{\mathcal{G}} \, \|v\|_{\mathcal{G}}. \tag{22}$$

Thus one index works for all such tests in the field unit ball, without a common bound on their divergence norms.

8.1 The original source channel

For each fixed clock value, the bounded Gaussian chain rule gives $D_B \sin(B(|W_t|)) = \cos(Z_t) v_{|W_t|}$ in the original source graph. One proves this by approximating the Brownian indicator with finite source projections and applying the closed Gaussian chain rule; $|\cos| \leq 1$ bounds the derivatives, and finite Gaussian moments justify the limit. Product Fubini gives the same graph on the joint space. The field $g_t = \cos(Z_t) v_{|W_t|}$ has continuous paths in H_B: the scenery and clock paths are continuous and $r \mapsto v_r$ is continuous in $L^2(\mathbb{R})$. Moreover

$$\|g_t\|_{H_B}^4 \leq |W_t|^2, \qquad \mathbb{E}\|g_t\|_{H_B}^4 \leq t.$$

For any convergent time sequence in a compact interval, these fourth moments are uniformly bounded. They imply uniform integrability of the squared field differences: their fourth moments are bounded by $8(\mathbb{E}\|g_{t_n}\|^4 + \mathbb{E}\|g_t\|^4)$, and the squared-difference tail above level R is at most this bound divided by R. Almost-sure continuity, truncation at R, and then $R \to \infty$ prove $g_{t_n} \to g_t$ in $L^2(P; H_B)$. This supplies strong time continuity, not an assumed pointwise evaluation map on L^2 classes.

Consequently the right Riemann sums (20) converge to the Bochner integral of g_t. The corresponding scalar Riemann sums converge in L^2 to U_T, since sin is Lipschitz and Z is mean-square continuous. Every finite pair is in the original source graph, and that graph is closed. Therefore

$$D_B U_T = \int_0^T \cos(Z_t) v_{|W_t|} \, dt, \qquad S_n(T) \to D_B U_T. \tag{23}$$

The bounded continuous-path observable U_T is measurable in the raw observed past at T, by its scalar Riemann sums.

8.2 The nonlinear covariance correction and a rough spatial operator

For the clock channel, set

$$Q_n(r) = \cos(B_n(r))B_n'(r), \qquad O_{nm}(r,z) = \int_{\mathbb{R}} \rho_n(r-x)\rho_m(z-x)\,dx.$$

The decisive bound is

$$|\mathbb{E}_B[Q_n(r)Q_m(z)]| \le O_{nm}(r,z) + 4. \tag{24}$$

It bounds the absolute expectation, not the expectation of the absolute product. In particular it does not assert independence between the values and derivatives of the same Gaussian scenery. Here is a direct derivation retaining those correlations. For any original Gaussian directions u, a, b, differentiation of their joint characteristic function gives

$$\mathbb{E}[X(a)X(b)\cos X(u)] = \big(\langle a,b\rangle - \langle u,a\rangle\langle u,b\rangle\big)e^{-\|u\|^2/2}.$$

Finite Gaussian second moments dominate the differentiated integrands, so this identity also includes degenerate directions. Use $\cos X(w)\cos X(v) = \frac{1}{2}[\cos X(w+v) + \cos X(w-v)]$. The covariance of the derivative directions is $O_{nm}(r,z)$. Every pairing of one mollified value direction with one derivative direction has absolute value at most one: the Brownian indicator pairing is an integral of a positive unit-mass mollifier over an oriented interval; averaging it against the other positive unit-mass mollifier preserves the bound. Thus the pairings with $w+v$ or $w-v$ are at most two, and each correction product at most four. The exponential factor is at most one. This proves (24) without a divergent derivative variance on the right.

Fix a finite interval $I = (a,b]$ of length $\ell = b-a \ge 0$ and define

$$J_n^I\phi = \int_I \phi(r)Q_n(r)\,dr \quad \text{as a Bochner integral in } L^2(P_B).$$

For fixed n the columns Q_n are strongly measurable and uniformly bounded in source L^2 by $\|\rho_n\|_2$. Hence every $\phi \in L^2(I)$ gives a well-defined integral. Since O_{nn} is symmetric, nonnegative and has whole-line row mass one, (24) and $2|\phi(r)\phi(z)| \le \phi(r)^2 + \phi(z)^2$ give, with all integrals on the finite product interval integrable,

$$\|J_n^I\phi\|^2_{L^2(B)} \le (1+4\ell)\|\phi\|^2_{L^2(I)}. \tag{25}$$

Indeed the row integral of $O_{nn}+4$ is at most $1+4\ell$; expansion of the squared Bochner norm and Fubini prove the estimate.

These operators converge strongly at every rough $L^2(I)$ input. To prove it, first take a smooth test ϕ. Ordinary pathwise integration by parts for the smooth B_n gives

$$J_n^I\phi = \phi(b)\sin B_n(b) - \phi(a)\sin B_n(a) - \int_a^b \phi'(r)\sin B_n(r)\,dr.$$

Canonical scenery convolution converges uniformly on compact spatial sets on the continuous-path event. The original Gaussian mollifier fields represent those convolutions on one event for all n, r: fixed rational-coordinate identities and continuity give this common event, without an uncountable null intersection. The displayed expression converges almost surely to the same formula with B in place of B_n, and is bounded by the deterministic constant $|\phi(a)| + |\phi(b)| + \int_a^b |\phi'|$. Dominated convergence therefore gives convergence in source L^2, including both endpoint terms. Smooth restrictions are dense in $L^2(I)$. The common bound (25) and a three-term approximation show $J_n^I\phi$ is Cauchy for every rough $\phi \in L^2(I)$. Its limit J^I is linear and bounded by $\sqrt{1+4\ell}$. This is pointwise strong operator convergence, not operator-norm convergence or a pointwise white-noise evaluation.

8.3 The summable occupation trace supplies full-vector convergence

For the actual linear occupation kernels of Theorem 7.1, an orthonormal basis (e_i) of the original clock coefficient space satisfies

$$\sum_i \|K_{e_i,T}\|^2_{L^2(W;L^2_x)} = \sum_i \|\langle D_W A_T, e_i\rangle\|^2_{L^2(P)} = \|D_W A_T\|_2^2 < \infty. \tag{26}$$

The first equality uses the original lifted scenery isometry; the second is Parseval followed by Tonelli. Extra unused source directions have zero contraction and cause no change.

Choose $0 \le \chi \le 1$ continuous with compact support inside I, and insert $\chi(|W_t|)$ in (19), obtaining $V_n^\chi(T)$. For every fixed clock direction h, the actual occupation identity gives

$$\langle V_n^\chi(T), h\rangle = J_n^I(\chi K_{h,T}) \quad \text{in } L^2(P_W; L^2(P_B)). \tag{27}$$

For precision, the right side applies the deterministic bounded operator to each clock-section input. To verify the identity, the $L^2(P_B)$-valued function $r \mapsto \chi(r)Q_n(r)$ is continuous and compactly supported. Test its spatial and clock-time integrals against any source Hilbert vector and apply the scalar occupation identity already proved. A countable dense set of such tests, plus Bochner integrability and separability, yields equality of the actual source-Hilbert vectors for almost every clock sample. Boundedness of the compactly supported column then gives their clock L^2 membership and product Fubini. This proves (27) on the original product law.

The strong convergence of J_n^I lifts through $L^2(P_W)$. For a fixed input $k(W)$, pointwise source convergence holds for almost every W, while the squared difference is bounded by $4(1 + 4\ell)\|k(W)\|^2$. Dominated convergence proves the lifted strong limit. In particular every contraction in (27) is Cauchy, and

$$\|\langle V_n^\chi(T), e_i\rangle\|_2^2 \le (1 + 4\ell)\|K_{e_i,T}\|^2_{L^2(W;L^2_x)}.$$

The bound is summable in i, uniformly in n, by (26). Parseval applied to $V_n^\chi - V_m^\chi$ now proves full-vector Cauchy convergence: each fixed coefficient difference tends to zero, its squared norm is bounded by four times this summable majorant, and a finite head plus summably small tail proves the sum tends to zero. This is the extra step that a collection of directional limits alone would lack.

8.4 Uniform spatial tails remove the cutoff

The cutoff error is uniform in the smoothing index. If $\chi = 1$ on $[-R, R]$, $R > 0$, then

$$\|V_n(T) - V_n^\chi(T)\|_2^2 \le \frac{T}{R^2} C_T. \tag{28}$$

We prove the dependence between the earlier clock and its tail is retained. Conditional on $W_s = x$, $s < t$, the folded later clock has density bounded by $2p_1(t - s)^{-1/2}$. Since the overlap row mass is one, any measurable $0 \le w \le 1$ gives

$$\mathbb{E}[w(W_s)O_{nm}(|W_s|, |W_t|)] \le 2p_1(t - s)^{-1/2}\mathbb{E}[w(W_s)].$$

This integrates the proved independent future Brownian increment, not an independent tail event. For $w(x) = 1_{|x|>R}$, Chebyshev and $\mathbb{E}W_s^2 = s$ give $\mathbb{E}w(W_s) \le s/R^2$. The nonlinear $+4$ term contributes at most $4s/R^2$ as well.

Expand the squared norm of the cutoff-error time integral. The coefficient inner product is $\langle q_s, q_t\rangle = \min(s, t)$; the signs have absolute value at most one. For $s < t$, the two nonnegative

complementary masks have product at most $1_{|W_s|>R}$. Combining the preceding weighted bound with (24), and using $s\min(s,t) \le T^2$, bounds the absolute time covariance by

$$\frac{T^2}{R^2}(2p_1|t-s|^{-1/2}+4).$$

The case $t < s$ follows by symmetry. The diagonal has two-dimensional Lebesgue measure zero, and $\int_0^T \int_0^T |t-s|^{-1/2}\,ds\,dt \le 4T\sqrt{T}$. Integration proves (28). With both masks replaced by one the same computation, using $\min(s,t) \le T$, gives $\|V_n(T)\|_2^2 \le C_T$. Choose compact cutoffs with radii tending to infinity. For any prescribed error, first make their uniform tail small; at that fixed cutoff use the full-vector Cauchy result. The triangle inequality therefore makes $V_n(T)$ Cauchy without cutoff. Completeness gives a strong limit $V(T)$ and (21) for that limit.

8.5 Identification in the original graph and the adjoint identity

At each width the original clock Gaussian chain rule gives $D_W \sin(B_n(|W_t|))$ equal to the integrand of (19). The kink at $W_t = 0$ causes no extra first weak derivative: $|\cdot|$ has almost-everywhere derivative sgn; smooth approximations give the closed-graph identity. More explicitly, condition on clock directions orthogonal to q_t and use the scalar coordinate $W_t/\sqrt{t}$ for $t > 0$. The mollified value has source second moment bounded by $(\sqrt{|r|}+\sqrt{\epsilon_n})^2$, and its spatial derivative has source second moment $\|\rho_n\|_2^2$, uniformly in r. These bounds are integrable against the scalar Gaussian clock density. Ordinary weak differentiation on bounded scalar intervals, followed by cutoff removal with these integrable bounds, proves the original Gaussian $W^{1,2}$ graph formula. Multiplying by the bounded cosine preserves the derivative bound. The graph, a closed linear subspace of value times field, is closed under Bochner integration of graph pairs. This follows, for example, by applying its bounded quotient map to the integral. Consequently

$$D_W \int_0^T \sin(B_n(|W_t|))dt = V_n(T).$$

The value error is at most $T\sqrt{\epsilon_n}$ by the sine Lipschitz bound and the original mollifier estimate. Its value tends to U_T and its field tends to $V(T)$ strongly. Closedness proves $U_T \in D(D_W)$ and $V(T) = D_W U_T$. Combined with (23) and observed measurability, this gives $U_T \in D(\mathcal{D})$ and both strong limits of the theorem. No weak convergence is promoted without the preceding summable-trace and tail proof.

The two channels in the approximating pair need not share one finite-stage graph input: the source channel is an unmollified finite-time quadrature, and the clock channel is a spatially mollified time integral. Their full-field limit is the actual joint gradient. The separate ramp-weighted cosine Simpson sums for U_T have their established value and adjoint-graph-dual limits; no strong native joint derivatives of those particular sums are inferred here. Finally the original adjoint identity gives $\langle \mathcal{D}U_T, v\rangle = \mathbb{E}[U_T \mathcal{D}^* v]$; its product is integrable by L^2 Cauchy–Schwarz. Subtracting and applying field Cauchy–Schwarz proves (22), completing the proof. This also permits tests depending on n with uniformly bounded field norms, even if their divergence norms grow. The horizon is fixed throughout. This nonlinear result supplies neither a positive-time point clock gradient nor an observed physical-time Itô or prescribed-driver representation theorem.

9 Formal scope and literature

The frozen authority is commit `bab688bb6140cd6316c80680095a914ea0d346db`. The accompanying theorem map records the exact average graphs, energy bounds, finite-average chain tests,

raw-past density, closed restrictions, maximal adjoints and time-limit declarations. Its central endpoint is `IBMPastAdjointTimeCalculus`. The full joint operator is `IBMJointIntrinsicDerivative`; its field channels are scenery and clock in that order. The additional nonlinear endpoints are `IBMSineAverageStrongClock` and `IBMSineAverageStrongJoint`; their summable coefficient trace and cutoff argument are reconstructed in full above. All scalar pairings above are on the actual joint law, with the Fubini factor order transported isometrically. The argument retains the original finite-energy domains. The weaker adjoint-graph dual is explicitly named and its norm is given; it is not a new native point gradient. The actual full-past adjoint can have a larger domain than the ambient adjoint restriction, and no equality of these domains has been assumed. Full baseline build and transitive axiom evidence were hash-verified; this editorial work creates no new Lean proof and does not repeat a resource-intensive root build. The Gaussian derivative/adjoint construction and closed-operator arguments are classical; see the Gaussian conventions in [2]. Allouba and Zheng [1] supply the Brownian-time model context. Nourdin and Zeineddine [3] study a signed Brownian clock in two-sided fractional scenery and an Itô-type formula, which is a different model and type of conclusion. The primary model definitions, Gaussian calculus conventions and relevant introductions were checked. Searches for average-domain and observed-past analogues were preliminary, not an exhaustive novelty audit. No independence, priority, or absence-of-precedent claim is made. The native-domain regularization, fixed-past maximal-adjoint statement and strong nonlinear approximation in Theorem 8.1 are the paper's precise verified content; full physical-time intrinsic calculus remains open.

References

[1] H. Allouba and W. Zheng, *Brownian-time processes: the PDE connection and the half-derivative generator*, Ann. Probab. 29 (2001), 1780–1795. arXiv:1005.3801.

[2] T. Sottinen and L. Viitasaari, *Stochastic analysis of Gaussian processes via Fredholm representation*, Int. J. Stoch. Anal. (2016), article 8694365. doi:10.1155/2016/8694365.

[3] I. Nourdin and R. Zeineddine, *An Itô's type formula for the fractional Brownian motion in Brownian time*, arXiv:1312.0818 (2013). Model and introduction comparison only.

[4] Project Stardust, *Gaussian clock-domain obstructions for reflected-clock Brownian motion*, reviewed editorial companion (2026), accompanying manuscript.

[5] J. Pitman and M. Yor, *A guide to Brownian motion and related stochastic processes*, primary author exposition (2018), occupation density discussion. `https://www.stat.berkeley.edu/~aldous/205B/pitman_yor_guide_bm.pdf`.